

Lecture 12: Reliability Under Distribution Shift

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1 Introduction

1.1 Standard ML

We first consider the standard supervised learning setting. Let P denote the source distribution over $\mathcal{X} \times \mathcal{Y}$. The training samples are drawn as

$$(x, y) \sim P.$$

A standard learning algorithm chooses a hypothesis by empirical risk minimization:

$$\min_{h \in \mathcal{H}} \mathbb{E}_{(x, y) \sim P} [\text{loss}(h(x), y)].$$

At test time, however, examples may be drawn from a different target distribution Q . The central question is how reliable the learned model remains when

$$P \neq Q.$$

1.2 Types of Distribution Shift

1. Covariate Shift

$$P(X) \neq Q(X), \quad \text{but } P(Y|X) = Q(Y|X).$$

e.g.: training on standard ImageNet images and testing on corrupted/stylized images.

2. Label Shift

Prior / class prevalence changes:

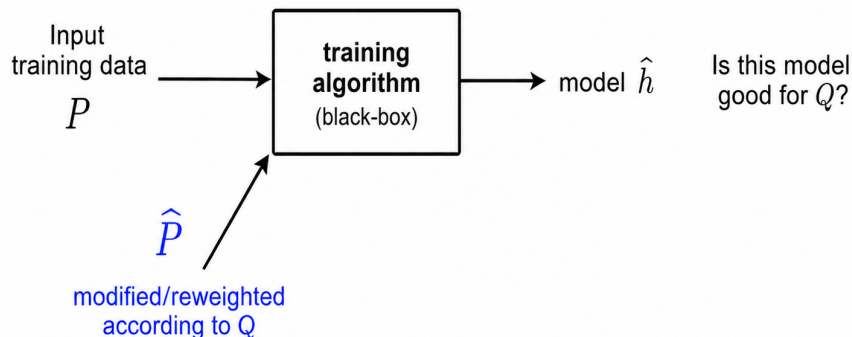
$$P(Y) \neq Q(Y), \quad P(X|Y) = Q(X|Y).$$

3. Concept Shift

$$P(Y|X) \neq Q(Y|X).$$

e.g.: the classification standards change.

Why Black-box? We treat the training algorithm as a black-box map from a training distribution to a learned model. The central question is whether a model trained on the source distribution P , or on a modified distribution $\hat{P}$, generalizes well to the target distribution Q .



1.3 Outline of Research Directions

1. (Unsupervised) Domain Adaptation: Generalization
2. Importance Sampling
3. Survey Calibration / Raking
4. Sinkhorn Scaling / Entropic Optimal Transport

The last two directions are closely connected and both lead to an **information geometry** perspective.

2 Domain Adaptation

In this section, we consider binary classification with 0-1 loss. Suppose a classifier is trained on the source distribution P . We want to bound its target risk under the distribution Q :

$$\text{loss}_Q(h) = \mathbb{E}_{(x,y) \sim Q} [\mathbf{1}\{h(x) \neq y\}].$$

Assume

$$h \in \mathcal{H} \quad (\text{classification}).$$

To relate source and target risks, we need a notion of discrepancy between the source and target marginal distributions that is relevant to the hypothesis class $\mathcal{H}$.

Definition 1 ($\mathcal{H}\Delta\mathcal{H}$ Divergence). For two distributions P and Q over $\mathcal{X}$, define

$$d_{\mathcal{H}\Delta\mathcal{H}}(P, Q) = \sup_{h, h' \in \mathcal{H}} \left| \Pr_{x \sim P_X} [h(x) \neq h'(x)] - \Pr_{x \sim Q_X} [h(x) \neq h'(x)] \right|.$$

For $D \in \{P, Q\}$, define the risk of a hypothesis h under distribution D as

$$\text{loss}_D(h) = \mathbb{E}_{(x, y) \sim D} [\mathbf{1}\{h(x) \neq y\}].$$

For two hypotheses $h, h' \in \mathcal{H}$, define their disagreement under D_X as

$$\text{disc}_D(h, h') = \Pr_{x \sim D_X} [h(x) \neq h'(x)].$$

Theorem 1 (Ben-David et al., 2010). For all $h \in \mathcal{H}$,

$$\text{loss}_Q(h) \leq \text{loss}_P(h) + d_{\mathcal{H}\Delta\mathcal{H}}(P, Q) + \lambda,$$

where

$$\lambda = \min_{h' \in \mathcal{H}} (\text{loss}_P(h') + \text{loss}_Q(h')).$$

Proof. Let

$$h^* \in \arg \min_{h' \in \mathcal{H}} (\text{loss}_P(h') + \text{loss}_Q(h')).$$

Then

$$\lambda = \text{loss}_P(h^*) + \text{loss}_Q(h^*).$$

We first use the triangle inequality for 0-1 loss. For any x ,

$$\mathbf{1}\{h(x) \neq y\} \leq \mathbf{1}\{h^*(x) \neq y\} + \mathbf{1}\{h(x) \neq h^*(x)\}.$$

Taking expectation over $(x, y) \sim Q$, we obtain

$$\text{loss}_Q(h) \leq \text{loss}_Q(h^*) + \text{disc}_Q(h, h^*).$$

Next, add and subtract the source-domain disagreement:

$$\text{disc}_Q(h, h^*) \leq \text{disc}_P(h, h^*) + |\text{disc}_Q(h, h^*) - \text{disc}_P(h, h^*)|.$$

By the definition of $d_{\mathcal{H}\Delta\mathcal{H}}(P, Q)$,

$$|\text{disc}_Q(h, h^*) - \text{disc}_P(h, h^*)| \leq d_{\mathcal{H}\Delta\mathcal{H}}(P, Q).$$

Therefore,

$$\text{loss}_Q(h) \leq \text{loss}_Q(h^*) + \text{disc}_P(h, h^*) + d_{\mathcal{H}\Delta\mathcal{H}}(P, Q).$$

Again by the triangle inequality for 0-1 loss,

$$\mathbf{1}\{h(x) \neq h^*(x)\} \leq \mathbf{1}\{h(x) \neq y\} + \mathbf{1}\{h^*(x) \neq y\}.$$

Taking expectation over $(x, y) \sim P$, we get

$$\text{disc}_P(h, h^*) \leq \text{loss}_P(h) + \text{loss}_P(h^*).$$

Substituting this into the previous inequality gives

$$\text{loss}_Q(h) \leq \text{loss}_Q(h^*) + \text{loss}_P(h) + \text{loss}_P(h^*) + d_{\mathcal{H}\Delta\mathcal{H}}(P, Q).$$

Rearranging,

$$\text{loss}_Q(h) \leq \text{loss}_P(h) + d_{\mathcal{H}\Delta\mathcal{H}}(P, Q) + (\text{loss}_P(h^*) + \text{loss}_Q(h^*)).$$

Since

$$\lambda = \text{loss}_P(h^*) + \text{loss}_Q(h^*),$$

we conclude that

$$\text{loss}_Q(h) \leq \text{loss}_P(h) + d_{\mathcal{H}\Delta\mathcal{H}}(P, Q) + \lambda.$$

□

3 Importance Sampling under Covariate Shift

Assume covariate shift:

$$P(Y|X) = Q(Y|X), \quad P_X \neq Q_X.$$

Target risk:

$$\text{loss}_Q(h) = \mathbb{E}_{(x,y) \sim Q} [\ell(h(x), y)] = \mathbb{E}_{x \sim Q_X} [\mathbb{E}_{y \sim Q(Y|X)} \ell(h(x), y)] = \mathbb{E}_{x \sim Q_X} [\mathbb{E}_{y \sim P(Y|X)} \ell(h(x), y)].$$

Rewriting the target risk using importance weights gives:

$$\text{loss}_Q(h) = \mathbb{E}_{x \sim P_X} \left[\frac{Q_X(x)}{P_X(x)} \mathbb{E}_{y \sim P(Y|X)} \ell(h(x), y) \right] = \mathbb{E}_{(x,y) \sim P} [w(x) \ell(h(x), y)].$$

where

$$w(x) = \frac{Q_X(x)}{P_X(x)}.$$

Training Procedure

Given:

- labeled samples from P (source),
- unlabeled samples from Q (target).

1. Estimate

$$w(x) = \frac{Q_X(x)}{P_X(x)}.$$

2. Learn h through weighted ERM.

3.1 Issues with Importance Sampling

1. The support condition

$$\text{supp}(Q_X) \subseteq \text{supp}(P_X)$$

is necessary. If the target distribution puts mass on regions never observed under the source distribution, the importance weight becomes infinite and source reweighting cannot recover the target risk.

2. How do we learn $w(x)$?

Statistically intractable in general (particularly in high dimensions).

Heuristic for Estimating $w(x)$

Reduce it to classification (e.g. logistic regression).

Use unlabeled data from P_X and Q_X .

1. Create a classification problem:

$$Z = +1 \quad \text{for } x \sim Q_X,$$

$$Z = -1 \quad \text{for } x \sim P_X.$$

2. Train a good classifier.

Assume we obtain the optimal classifier.

Using Bayes rule:

$$\Pr(Z = 1|X = x) = \frac{Q_X(x) \Pr(Z = 1)}{Q_X(x) \Pr(Z = 1) + P_X(x) \Pr(Z = -1)}.$$

Hence,

$$w(x) = \frac{Q_X(x)}{P_X(x)} = \frac{\Pr(Z = 1|x)}{1 - \Pr(Z = 1|x)} \cdot \frac{N_P}{N_Q}.$$

Here N_P and N_Q denote the numbers of unlabeled samples drawn from the source marginal P_X and target marginal Q_X , respectively; the factor N_P/N_Q corrects for the empirical class prior in the domain-classification problem.

References

- [1] Shai Ben-David, John Blitzer, Koby Crammer, Alex Kulesza, Fernando Pereira, and Jennifer Wortman Vaughan. A theory of learning from different domains. *Machine Learning*, 79:151–175, 2010. doi:10.1007/s10994-009-5152-4.